

Agnes 2B: A Reproducible and Cost-Aware Two-Stage Pretraining Protocol on Consumer GPU Clusters

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Abstract

We present Agnes 2B, a reproducible pretraining protocol for a compact dense language model on consumer-grade GPU clusters. The proposed model contains approximately 2.032 billion parameters, uses 28 decoder blocks and a 4,096-token training context, and is trained on 1.3988 trillion tokens in two stages. The data schedule moves from broad language coverage toward greater emphasis on mathematics, code, and instruction-formatted data while retaining controlled replay. The training method combines blockwise E4M3 FP8 linear operations, BF16/FP32 computation for numerically sensitive components, a hybrid MuonH–AdamW optimizer, two-stage learning-rate scheduling, source-local curriculum ordering, and late-stage checkpoint weight averaging. The distributed configuration scales from 24 to 96 RTX 5090 GPUs using tensor parallelism of one and topology-aware pipeline and data parallelism. Based on measurements from a structurally matched reference system, the full-scale run is projected to require 22,514 active GPU-hours and 17.59 days of pure training time. The capability plan uses reference-reported endpoints of 43.50, 63.02, and 57.81 for Math/Code Avg4, Reason/Knowledge Avg11, and Avg15, while Agnes preregisters release floors of 42.0, 62.5, and 57.0, together with final validation loss at most 2.52. We further specify controlled proxy studies, ablations, evaluation metrics, recovery requirements, and preregistered acceptance criteria. This paper defines a falsifiable training protocol rather than reporting completed Agnes training results.

Keywords: foundation-model pretraining; consumer GPU clusters; mixed-precision training; curriculum learning; distributed training; reproducibility.

Evidence convention. **Reported evidence** identifies measurements from prior work; **Agnes design** identifies a proposed Agnes configuration; and **Release gate** identifies a threshold that must be measured before scale-up or release. Unless stated otherwise, cost denotes accelerator-equivalent cost rather than total project expenditure.

Table 1. Core specification of the proposed Agnes 2B program.

Item	Specified value	Item	Specified value
Model scale	~2.032B, dense	Training tokens	1.3988T
Training context	4,096	Global batch size	1,536 sequences
Phase 1	24 GPUs, 438.84B	Phase 2	96 GPUs, 959.99B
Primary arithmetic	Blockwise E4M3 FP8	Persistent weights	BF16
Production GPU-hours	22,514, planning anchor	Pure training time	17.59 days, planning anchor
Primary evaluation	15-task suite plus Chinese evaluation	Final performance floor	Loss ≤ 2.52 ; Avg15 ≥ 57.0 ; Avg4 ≥ 42.0 ; Avg11 ≥ 62.5

1 Introduction

1.1 Motivation and Objectives

Agnes 2B is intended to be a locally deployable, continually trainable, and auditable general-purpose base model. The pretraining program pursues four concurrent objectives: stable English and Chinese knowledge coverage; a transferable foundation for mathematical reasoning and code generation; reproducible throughput on consumer GPU clusters; and complete freezing of data manifests, sample order, training state, and evaluation protocols so that subsequent supervised fine-tuning, preference optimization, and agent training can be attributed to a well-defined base checkpoint.

Success is not defined merely as processing 1.4T tokens. A successful run must satisfy all of the following within the approved budget: no unrecoverable optimization divergence; traceable data lineage and licensing; validation loss within a prespecified scaling envelope; core capability above release thresholds; and a final checkpoint whose tensor shapes, parameter grouping, hashes, and recovery behavior can be reproduced from the released configuration.

1.2 Scope

The report covers tokenizer construction, data acquisition and governance, model architecture, low-precision numerical execution, optimization, curriculum ordering, parallelism, failure recovery, base-model evaluation, and the release package. It does not cover supervised fine-tuning, preference optimization, reinforcement learning, tool-use policy, or product-level safety alignment; those activities begin from the Base checkpoint produced here and require separate experimental plans.

The numerical budget applies only to the fixed dense architecture described in Section 3. It must not be extrapolated directly to 7B, 120B, or mixture-of-experts models. The primary hardware target is an eight-GPU node with 32GB RTX 5090 devices. Alternative accelerators require new throughput, memory, communication, and wall-clock measurements.

1.3 Contributions and Non-Claims

This report makes four practical contributions. First, it converts favorable results from a low-cost pretraining study [1] into explicit, testable design hypotheses. Second, it integrates model, data, optimizer, arithmetic, and systems decisions into a single token-accounted training plan. Third, it defines gates that can stop the project before the high-cost production run. Fourth, it specifies reproducibility artifacts and evaluation rules before results are observed.

No performance number in this report is an Agnes result unless it is explicitly labeled as such in a future revision. Values inherited from the reference study remain reported evidence; values such as architecture settings, budgets, and thresholds are proposed decisions or release criteria. This distinction prevents planning targets from being presented as empirical achievements.

2 Design Rationale and Evidence Mapping

2.1 Evidence-to-Decision Mapping

Table 2 summarizes how each major observation reported in the reference study [1] is converted into an Agnes design choice and an independent pre-run test.

Table 2. Reported evidence, Agnes adoption, and required revalidation.

Reported evidence	Agnes design choice	Validation before full training
FP8-BF16 validation-loss difference of 0.0031–0.0039 and 1.36× throughput at the 1.7B proxy scale	Use blockwise E4M3 from random initialization while retaining sensitive operators in BF16/FP32	Paired 170M and 600M runs with identical data and seeds; loss difference ≤ 0.005 and throughput $\geq 1.25\times$
Approximately 1.19× quality-adjusted compute gain from the complete MuonH recipe, based on short scaling runs	Apply MuonH only to approximately scale-invariant attention and MLP matrices; optimize all other parameters with AdamW	At least two model scales and two seeds; compare against tuned AdamW and standard-Muon baselines
Approximately 1.18 points of average-score improvement from source-local curriculum ordering at a single endpoint	In Phase 2, order examples from lower to higher preference within each source; never compare raw scores across sources	Train a uniform-order shadow branch on the identical token multiset and interpret any gain only after contamination analysis
Production configurations of 24 and 96 GPUs sustained 238 and 192 TFLOP/s/GPU, respectively	Use TP=1 with communication-aware pipeline and data parallelism, including asymmetric layer placement	Run a 1,000-step systems trial on supported drivers; recompute time and cost if throughput gates are missed
USD 6.89K covers active GPU time after data materialization only	Treat this figure as a production-accelerator planning anchor, never as the total project price	Account separately for data, experiments, storage, evaluation, labor, compliance, and rerun capacity

Boundary of the evidence

The reference study [1] does not disclose the production tokenizer, every random seed, the complete AdamW configuration, or several numerical-stability settings. Most efficiency conclusions were obtained from scaling runs near 20 tokens per parameter, whereas the proposed production run is close to 700 tokens per parameter. Agnes therefore treats the reported methods as high-value starting hypotheses, not as results that transfer without verification.

2.2 Two-Stage Program Logic

The training program separates broad distributional coverage from targeted capability development. Phase 1 establishes a robust language-model prior over English, Chinese, mathematics, and code. Phase 2 increases the representation of mathematics, code, and instruction-like material; introduces a smooth replay transition; imposes source-local curriculum order; and concludes with a constant-learning-rate branch followed by checkpoint averaging. This staging allows the team to isolate

distribution shifts, audit replay accounting, and retain fallback endpoints if the curriculum or late-stage averaging does not generalize.

3 Model and Tokenizer Design

3.1 Architecture

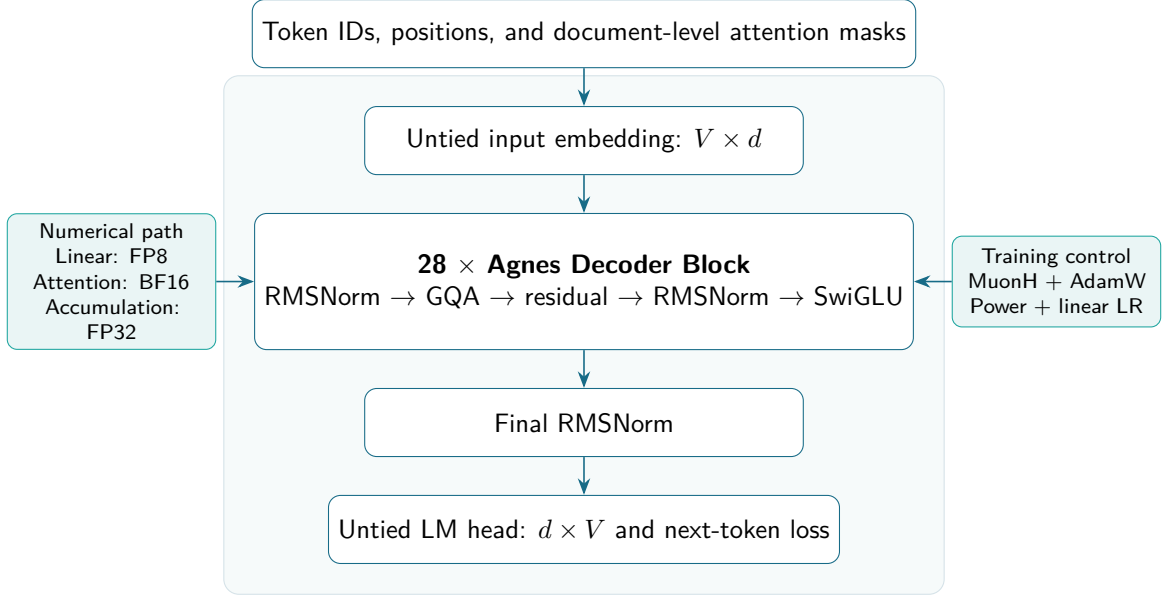


Figure 1. Information flow and precision boundaries in Agnes 2B. The input embedding and output head do not share weights.

Table 3. Proposed Agnes 2B architecture.

Configuration item	Value	Rationale or constraint
Model family	Dense decoder-only	Causal language modeling only
Layers / hidden size	28 / 2,048	Compatible with a mature 1.7B-class backbone configuration [2]
FFN intermediate size	6,144	SwiGLU with three bias-free projections
Query / KV heads	16 / 8	Grouped-query attention; head dimension 128
Vocabulary size	151,936	Includes 256 reserved control tokens
Embedding / LM head	Untied	Approximately 311.16M parameters on each side; pipeline placement must account for both
Normalization	Pre-RMSNorm	$\epsilon = 10^{-6}$; optional head-wise RMSNorm on Q/K
Positional encoding	RoPE	$\theta = 10^6$; train at 4,096 and preserve structural capacity to 40,960
Activation / dropout	SiLU / 0.0	Overfitting is addressed through data scale and deduplication
Total parameters	Approximately 2.032B	Final tensor-by-tensor enumeration is authoritative

Ignoring small normalization terms, the parameter count is approximated by

$$N \approx 2Vd + L [2d^2 + 2d(H_{kv}d_h) + 3dd_{ff}]. \quad (1)$$

With $V = 151,936$, $d = 2,048$, $L = 28$, $H_{kv} = 8$, $d_h = 128$, and $d_{ff} = 6,144$, Eq. (1) yields approximately 2.03B parameters. The configuration gate nevertheless requires exact agreement

among the exported parameter inventory, optimizer-group membership, and checkpoint tensor shapes.

3.2 Training Objective, Initialization, and Packing

The objective is standard next-token cross-entropy,

$$\mathcal{L}(\theta) = -\frac{1}{\sum_i m_i} \sum_i m_i \log p_\theta(x_i | x_{<i}), \quad (2)$$

where m_i masks padding, invalid positions after document boundaries, and control tokens excluded from the loss. Length-aware packing is used for throughput. Documents within one packed sequence receive a block-diagonal causal mask and independently reset position indices, preventing artificial cross-document adjacency from becoming a learned regularity.

Linear layers use a truncated-normal initialization with baseline standard deviation 0.02. Residual-branch output projections are additionally scaled by $1/\sqrt{2L}$. Before scale-up, the team must record layerwise activation standard deviation, maximum absolute value, and FP8 saturation immediately after initialization. Any non-finite value blocks progression.

3.3 Tokenizer Construction

Because the reference study [1] does not specify its production tokenizer, Agnes treats tokenizer development as a separate workflow that must finish before token budgets are frozen:

1. Sample approximately 100GB of text at document level from the final licensing whitelist, following the target proportions for English, Chinese, mathematics, and code.
2. Train a byte-level BPE tokenizer with a total vocabulary of 151,936 tokens and byte fallback, guaranteeing lossless round-trip behavior for arbitrary UTF-8 input.
3. Apply only UTF-8 validation and NFC normalization to natural language. Preserve indentation, whitespace, and case in code; normalize line endings only.
4. Reserve 256 fixed token IDs inside the total vocabulary for BOS/EOS/PAD, fill-in-the-middle, conversation boundaries, tools, and future multimodal placeholders. Inactive tokens are not inserted into pretraining text.
5. Measure fertility, unknown-byte rate, and round-trip fidelity separately for all four domains. Relative to the frozen comparison tokenizer, English, Chinese, and code fertility may not worsen by more than 3%, and round-trip fidelity must be 100%.

Tokenizer freeze gate T0

Training-token ledgers may be recomputed only after the tokenizer model, normalizer, special-token map, training-sample manifest, random seed, and SHA-256 hashes are frozen. Any subsequent tokenizer change invalidates prior token counts and step estimates and therefore requires rematerializing all shards.

4 Data Mixture, Governance, and Curriculum

4.1 Domain Allocation Across Two Phases

Phase 1 prioritizes broad language-model coverage. Phase 2 increases the relative weight of mathematics, code, and instruction-like material while using replay to avoid an abrupt distribution shift. Table 4 is the initial materialization target. Proxy results may motivate changes within a domain, but the phase-level token ledger must remain fixed after data gate D0.

Table 4. Domain-level token budgets for the two training phases.

Domain	Phase 1 tokens	Phase 1 share	Phase 2 pool	Phase 2 share
General English and knowledge	321.31B	73.2%	557.52B	59.4%
Mathematics and formal reasoning	31.56B	7.2%	171.29B	18.3%
General Chinese and knowledge	51.34B	11.7%	88.48B	9.4%
Code and software engineering	34.62B	7.9%	108.11B	11.5%
Instruction-like data	0	0.0%	12.66B	1.3%
Component-pool total	438.84B	100.0%	938.06B	100.0%

Displayed components are rounded to 0.01B tokens and shares to 0.1 percentage points; the phase totals are authoritative.

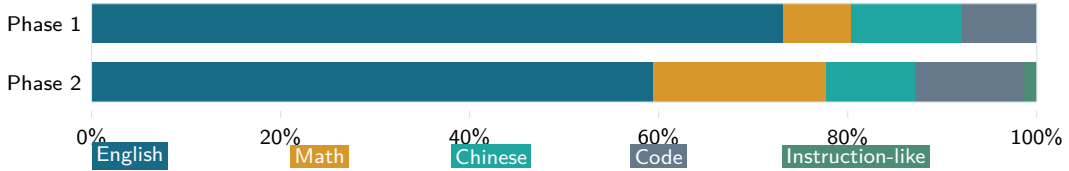


Figure 2. Domain mixture by phase. Phase 2 assigns substantially more capacity to mathematics and code. Bars are normalized to 100% after rounding.

4.2 Sources and Governance Pipeline

Candidate sources should prioritize traceable open-web educational material, encyclopedic and academic text, clearly licensed code, mathematical reasoning and proof data, Chinese knowledge corpora, and limited quantities of tool-use, long-context, and terminal trajectories. A dataset name is not itself a licensing conclusion. Every version must record its publisher, original URL, version, license, redistribution restrictions, jurisdictional constraints, access date, and deletion-request mechanism.

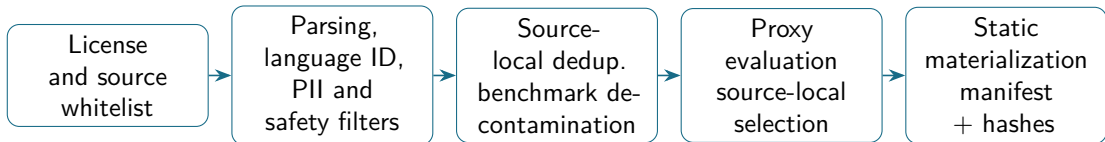


Figure 3. Agnes data-governance pipeline. Production training uses neither online scoring nor dynamically changing mixtures.

The immutable processing order is license admission → document parsing → language and domain classification → PII, malicious-code, and low-quality filtering → exact and MinHash deduplication → benchmark decontamination → tokenization and packing → static shards. Ordinary web corpora are first deduplicated within source. Benchmark questions, mirrored code repositories, and heavily reused synthetic data additionally undergo cross-source near-duplicate detection. Every shard records source-document IDs, the transformation chain, token counts, random seeds, and hashes.

4.3 Proxy-Based Data Selection

Agnes uses a shared 0.6B proxy checkpoint to measure the capability direction induced by each candidate source; raw quality scores are never compared across unrelated sources. The proposed protocol first trains a common starting point on 86B Phase 1 candidate tokens. Each candidate condition then continues for approximately 8.4B tokens: 2,000 steps at sequence length 4,096 and global batch size 1,024. The candidate-source share rises linearly from 0 to 80% over the first 1,600 steps and remains at 80% for the final 400 steps.

For scored sources larger than 50B tokens, the experiment evaluates the source-local q00, q25, q50, and q75 slices. Sources containing 5B–50B tokens, or sources without a reliable score, contribute a fixed sample of approximately 4B tokens. Each condition is reported as a four-axis capability vector covering mathematics, code, Chinese, and general knowledge, with all task-level scores retained. Principal component analysis may summarize capability direction but is not treated as causal evidence. Selections among the five largest Phase 2 sources require two sample-order seeds. When observed differences are smaller than evaluation noise, selection defaults to licensing clarity, unique-token contribution, and processing cost.

4.4 Phase 2 Transition and Source-Local Curriculum

The Phase 2 component pool contains 938.06B tokens. An additional 21.93B tokens are replayed from Phase 1, producing 959.99B physical Phase 2 tokens. The first 43.87B tokens form a linear transition: the Phase 1 replay weight decreases from 100% to zero while the Phase 2 weight increases from zero to 100%. Consequently, approximately 21.93B tokens come from replay and approximately 21.93B come from the beginning of the Phase 2 pool. The latter quantity must not be counted twice.

Following the curriculum-learning motivation in prior work [6], curriculum order is defined only within each source. Components with a validated score are arranged from lower to higher preference; components without a score use a fixed random permutation. Cumulative token mass is mapped to 376 progress buckets, each containing approximately 2.5B tokens. Each bucket draws the corresponding percentile interval from all components, approximately preserving the mixture across sources. A uniform control branch uses the identical token multiset but applies a global random shuffle.

Data freeze gate D0

Phase 1, the Phase 2 pool, and replay each require three independent accounting views: a unique-document ledger, a materialized-token ledger, and a physical-training-token counter. They must never be conflated. The published budget is defined by the materialized-token ledger, while the running non-padding token counter provides real-time reconciliation. A discrepancy above 0.1% automatically pauses training.

5 Optimization and Numerical Precision

5.1 Grouped Optimizers

Following the constrained matrix-optimization formulation of Wen et al. [5], MuonH updates only explicitly whitelisted matrices in the attention and MLP modules. Embeddings, RMSNorm parameters, the LM head, biases, and all remaining tensors are optimized with AdamW. The MuonH group uses zero weight decay and ten times the base learning-rate schedule. The AdamW group uses weight decay 0.1, with proposed initialization values $\beta_1 = 0.9$, $\beta_2 = 0.95$, and $\epsilon = 10^{-8}$. These AdamW settings are Agnes engineering priors and require validation at 170M and 600M scales.

For a selected matrix W_t , let $R = \|W_0\|_F$ and normalize the Muon update u_t as $\hat{u}_t = u_t / \|u_t\|_F$. The constrained update is

$$\widetilde{W}_{t+1} = W_t - \eta_t R \hat{u}_t, \quad W_{t+1} = R \frac{\widetilde{W}_{t+1}}{\|\widetilde{W}_{t+1}\|_F}. \quad (3)$$

This update preserves matrix radius and makes η_t directly interpretable relative to weight scale. The implementation must log weight norm, update norm, and effective learning rate for every parameter group. A matrix that repeatedly triggers extreme projection or anomalous norms is removed from the MuonH whitelist and retested from the most recent healthy checkpoint at proxy scale.

5.2 FP8 Mixed-Precision Path

Table 5. Proposed low-precision numerical path.

Operator or state	Precision	Rule
Q/K/V, attention-output, and MLP projections	E4M3 FP8 GEMM	Forward, data-gradient, and weight-gradient GEMMs use blockwise execution
Activation / activation gradient	FP8, 1×128 group	Online scaling along the GEMM reduction dimension
Weight quantization	FP8, 128×128 block	Scales restricted to powers of two; saturation is logged
Core attention, residual path, RMSNorm	BF16	Sensitive reductions remain outside FP8
Softmax, gradient reduction, optimizer accumulation	FP32	Master weights and optimizer states retain high precision
Checkpoint weights	BF16	Deployment quantization is a separate post-training stage

The implementation follows blockwise FP8 execution guidance [4]. FP8 is enabled from random initialization rather than after a fixed BF16 warmup, while per-layer BF16 fallback remains available. Every 50 steps, the system records non-finite values, amax statistics, scales, saturation, gradient norms, and loss spikes. Training pauses automatically if any critical layer exceeds 0.5% FP8 saturation for three consecutive observations, if paired validation loss differs by more than 0.005, or if a non-finite gradient appears. Recovery begins from the preceding healthy checkpoint with the implicated layers moved to BF16.

5.3 Two-Stage Learning-Rate Schedule

The maximum number of physical tokens per optimization step is

$$B_{\text{tok}} = 1,536 \times 4,096 = 6,291,456. \quad (4)$$

Phase 1 therefore requires approximately 69,752 steps. The base learning rate warms up linearly for 1,000 steps and then follows an open-ended power schedule motivated by token-agnostic scheduling work [7]:

$$\eta_{\text{base}}(k) = \begin{cases} 5 \times 10^{-3} k / 1000, & 0 \leq k \leq 1000, \\ 5 \times 10^{-4} + 4.5 \times 10^{-3} \left(1 + \frac{k - 1000}{1000}\right)^{-1/2}, & k > 1000. \end{cases} \quad (5)$$

The terminal Phase 1 base rate is approximately 1.04×10^{-3} . Phase 2 contains approximately 152,586 steps and linearly decays the main trajectory from 1.04×10^{-3} to 10^{-5} :

$$\eta_{\text{base}}(s) = \eta_0 + (\eta_{\text{min}} - \eta_0) \frac{s}{S_2}. \quad (6)$$

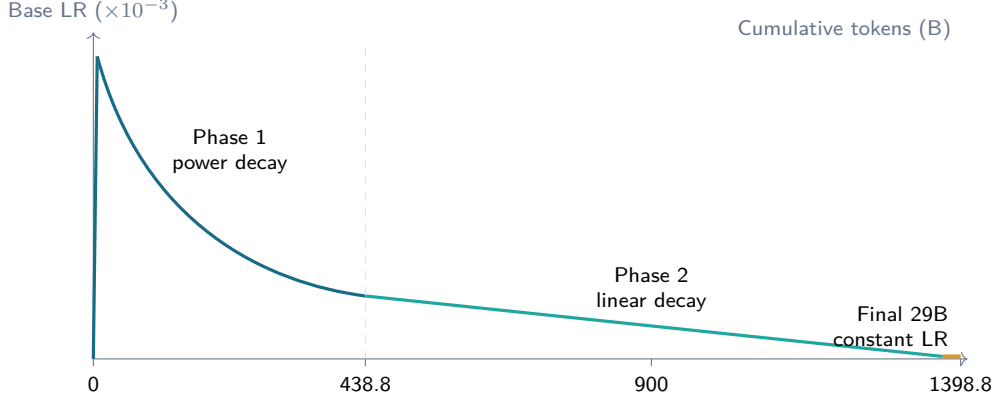


Figure 4. Proposed base learning-rate schedule. The MuonH group uses ten times the plotted values.

5.4 Constant-Learning-Rate Branch and Model Averaging

For approximately the final 29B Phase 2 tokens, training resumes from a late checkpoint selected by the physical non-padding token counter. The base learning rate is held near 4.08×10^{-5} , giving a MuonH effective rate of 4.08×10^{-4} . Model weights are saved every 100 steps over approximately the final 3.15B tokens. Six terminal checkpoints are averaged with equal weight:

$$\bar{\theta} = \frac{1}{6} \sum_{j=1}^6 \theta_j. \quad (7)$$

Only model parameters are averaged; optimizer state, scheduler state, and data cursors are not. Agnes does not hard-code global step identifiers from the reference study. The branch point and averaging window are defined by frozen token ledgers and the non-padding token counter, avoiding discrepancies caused by padding, partial batches, or incompatible counting conventions.

6 Training System and Cluster Topology

6.1 Node and Software Baseline

Each node contains eight 32GB RTX 5090 GPUs, with four GPUs associated with each CPU socket, dual-socket x86 processors, at least 512GB of system memory, at least 8TB of local NVMe storage, and one dual-port ConnectX-7 network adapter. Inter-node communication uses 400Gbps InfiniBand. The production manifest freezes the exact Linux, NVIDIA driver, CUDA, NCCL, Megatron Core [3], and Transformer Engine [4] versions together with the container digest.

Production runs use vendor-supported drivers and communication paths by default. The reference implementation relied on unsupported P2P/GDR modifications that cannot be disclosed or reproduced completely; Agnes therefore excludes them from the reproducibility baseline. Any experimental communication optimization must first pass compliance, security, and operations review and must retain a stock-driver baseline, a rollback script, and a separately labeled throughput report.

6.2 Parallelism and Pipeline Balance

Because RTX 5090 devices lack NVLink, tensor parallelism would introduce expensive collectives at every layer. Both phases therefore use TP=1 and combine pipeline parallelism with data parallelism, as shown in Table 6.

Table 6. Production parallelism configurations and system gates.

Phase	GPUs	TP / PP / DP	Pipeline layers	Reported rate	Agnes gate
Phase 1	24	1 / 2 / 12	(18 10)	238 TFLOP/s/GPU	≥ 220 TFLOP/s/GPU
Phase 2	96	1 / 4 / 24	(9 9 9 1)	192 TFLOP/s/GPU	≥ 175 TFLOP/s/GPU

The pipeline uses a 1F1B schedule and **pp-dp** rank ordering so that ranks in one pipeline group remain topologically close where possible. Micro-batch size is 2. Because the untied LM head contains approximately 311M parameters and is not concealed by FP8 execution of the internal layers, the last stage receives only one Transformer layer. MuonH tensors remain whole where possible; Adam state is distributed to balance residual memory. Stage execution times must differ by less than 8%, and every stage must retain at least 1.5GB of peak-memory headroom.

6.3 Checkpointing, Recovery, and Observability

The fault-tolerance protocol is as follows:

- Save a fully recoverable state every 1,000 steps, including model parameters, master weights, optimizer, scheduler, random-number generators, data cursor, non-padding token counter, and numerical scalars. Save lightweight model weights every 250 steps.
- Retain the three latest full checkpoints locally and two remote replicas. After remote object creation, verify file-level SHA-256 hashes and reread a random sample of tensors.
- Record loss, gradient norm, effective learning rate, throughput, memory use, temperature, and communication time every 10 steps; evaluate the fixed validation set every 500 steps.
- After node loss, restore the latest full state and compare loss and throughput against the historical band for 100 steps. Roll back if deviation exceeds three standard deviations over the matched historical window.
- Conduct a recovery drill every 24 hours. The recovery-point objective is at most 1,000 steps and the recovery-time objective is at most two hours.

7 Compute, Cost, and Execution Schedule

7.1 Compute and Wall-Clock Estimate

For a dense Transformer with approximately 2.032B parameters and 1.3988T training tokens, the customary order-of-magnitude estimate is

$$C \approx 6ND \approx 1.71 \times 10^{22} \text{ FLOPs.} \quad (8)$$

The $6ND$ approximation is a consistency check rather than a substitute for framework-level FLOP accounting. Measurements reported for the structurally matched production setup imply 6,009 GPU-hours and 10.43 days for Phase 1, followed by 16,505 GPU-hours and 7.16 days for Phase 2. The aggregate is 22,514 GPU-hours and 17.59 days. Agnes adds a 15% execution allowance, yielding approximately 20.2 days of pure training capacity, plus two to three days for stage gates, recovery drills, and final averaging.

Table 7. Production-training planning baseline. Values are not Agnes measurements.

Phase	GPUs	Tokens	Wall time	GPU-hours	Equivalent cost
Phase 1	24	438.84B	10.43 days	6,009	USD 1.84K
Phase 2	96	959.99B	7.16 days	16,505	USD 5.05K
Production total	–	1.3988T	17.59 days	22,514	USD 6.89K
15% rerun allowance	–	–	2.64 equivalent days	3,377	USD 1.03K

The accounting rate is approximately USD 0.306 per GPU-hour under a five-year depreciation-plus-electricity convention; it is not a public-cloud market rate. A complete project budget must additionally cover CPU preprocessing, data storage, proxy and scaling experiments, failed trials, networking and facility costs, evaluation, release engineering, compliance, and personnel.

7.2 Accelerator Budget Envelope

Table 8. Proposed accelerator budget envelope, excluding labor and infrastructure procurement.

Budget item	Allowance	Accounting scope
Production run	USD 6.89K	Reference-system measurement; recalculate after the Agnes 1,000-step systems trial
Production rerun allowance	USD 1.03K	15% of the production run for node failures and one local fallback
Tokenizer, proxy, and data ablations	USD 1.0K–1.8K	Determined by the number of conditions, seeds, and proxy tokens
Precision, optimizer, and system scaling	USD 1.0K–1.7K	Selective 170M/330M/600M/1.09B/1.7B experiments
Recommended accelerator envelope	USD 9.9K–11.4K	Excludes data storage, networking, power equipment, infrastructure purchase, and labor

7.3 Eight-Week Execution Plan

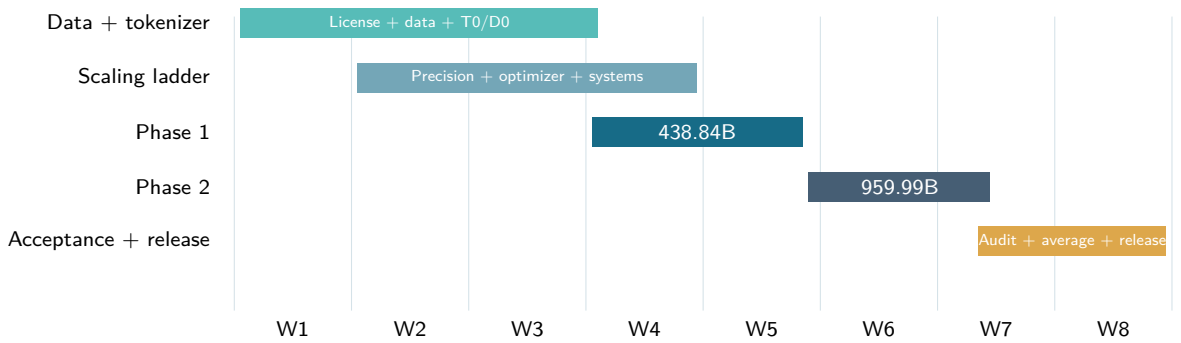


Figure 5. Indicative eight-week schedule. Data work and system scaling overlap; the production run begins only after all prerequisite gates pass. Bars denote planning windows, not measured completion dates.

8 Evaluation Protocol and Release Gates

8.1 Fixed Evaluation Protocol

The primary base-capability suite contains 15 tasks: GSM8K, MATH, sanitized-MBPP, HumanEval, MMLU, MMLU-Pro, ARC-Challenge, ARC-Easy, BoolQ, CommonsenseQA, HellaSwag, PIQA, SocialIQA, WinoGrande, and BBH. Generative tasks use greedy decoding and a frozen parser. Multiple-choice tasks report both perplexity-based and generated-answer metrics; the primary convention is frozen before training and may not be selected retrospectively per model.

Chinese evaluation adds C-Eval, CMMLU, and an internal decontaminated Chinese knowledge-and-reasoning set. Code evaluation adds multilingual unit tests. At the Base-model stage, long-context testing is limited to consistency within 4K tokens. Results obtained by extrapolating RoPE beyond the trained context do not contribute to the primary release score. Every benchmark freezes its version, prompt template, shot count, maximum generation length, stop sequences, parser, and container digest.

The aggregate labels *Avg15*, *Math/Code Avg4*, and *Reason/Knowledge Avg11* refer to metric groups frozen in the evaluation manifest before the first production checkpoint is evaluated. The manifest must enumerate group membership and normalization rules; the labels alone are insufficient to redefine a group after scores are observed.

8.2 Reference Performance and Agnes Target Profile

Table 9 records the final base-model scores reported by the reference study under its common OpenCompass pipeline [1]. These percentages are **external performance anchors**: they are neither Agnes measurements nor forecasts. Individual task values remain diagnostic because small evaluation or tokenizer changes can move a single score materially. Agnes preregisters hard floors only for validation loss and the three aggregates, with stretch goals used to rank checkpoints that already pass every release gate.

The reference pipeline uses greedy generation for GSM8K, MATH, sanitized-MBPP, HumanEval, MMLU-Pro, and BBH, and perplexity ranking for the other nine tasks. For applicable multiple-choice tasks, the reference aggregate selects the better cloze or multiple-choice formulation independently for each model. Agnes instead freezes one primary formulation before training. Consequently, a claim of numerical parity requires a protocol-matched rerun; the tabulated values alone do not establish parity.

Table 9. Reference-reported performance anchors and their preregistered Agnes use. All capability scores are percentages.

Indicator	Reference protocol	Reported anchor	Agnes preregistered use
Phase 1 validation loss	Held-out next-token loss	2.730	G4 floor ≤ 2.75 ; stretch ≤ 2.73
Phase 2 trajectory loss	Held-out next-token loss	2.488	G5 candidate floor ≤ 2.52 ; measure the averaged export separately
GSM8K	GEN, 4-shot	59.67	Diagnostic only; no task-level floor
MATH	GEN, 4-shot	30.30	Diagnostic only; no task-level floor
sanitized-MBPP	GEN, 3-shot	52.92	Diagnostic only; no task-level floor
HumanEval	GEN, 3-shot	31.10	Diagnostic only; no task-level floor

Table 9. Reference-reported performance anchors and their preregistered Agnes use (continued).

Indicator	Reference protocol	Reported anchor	Agnes preregistered use
Math/Code Avg4	Unweighted mean of four tasks	43.50	G5 floor ≥ 42.0 ; stretch ≥ 43.5
MMLU	PPL, 5-shot	57.44	Diagnostic only; no task-level floor
MMLU-Pro	GEN, 5-shot CoT	25.81	Diagnostic only; no task-level floor
ARC-Challenge	PPL, 5-shot	74.24	Diagnostic only; no task-level floor
ARC-Easy	PPL, 5-shot	86.07	Diagnostic only; no task-level floor
BoolQ	PPL, 5-shot	76.85	Diagnostic only; no task-level floor
CommonsenseQA	PPL, 5-shot	67.81	Diagnostic only; no task-level floor
HellaSwag	PPL, 5-shot	63.21	Diagnostic only; no task-level floor
PIQA	PPL, 5-shot	75.52	Diagnostic only; no task-level floor
SocialQA	PPL, 5-shot	63.61	Diagnostic only; no task-level floor
WinoGrande	PPL, 5-shot	60.62	Diagnostic only; no task-level floor
BBH	GEN, 4-shot	42.05	Diagnostic only; no task-level floor
Reason/Knowledge Avg11	Unweighted mean of eleven tasks	63.02	G5 floor ≥ 62.5 ; stretch ≥ 63.0
Avg15	Unweighted mean of all tasks	57.81	G5 floor ≥ 57.0 ; stretch ≥ 57.8
Chinese extension	C-Eval, CMMLU, internal set	—	Freeze numeric thresholds after 600M calibration; report separately from Avg15

The reference study does not report Chinese-specific base-model scores. Agnes therefore does not invent an absolute Chinese target from unrelated evidence. The 600M calibration run must freeze the C-Eval, CMMLU, and internal-set thresholds before the 1.7B scaling run, while G4 retains a no-material-regression condition throughout production.

8.3 Stage Gates

Table 10. Agnes pretraining and release gates.

Gate	Time	Pass criterion	Action on failure
G0	Data and pipeline	Tokenizer and shard hashes frozen; 100-step replay loss difference $< 10^{-5}$; no exact benchmark-data match under the frozen contamination protocol	Block scale-up and roll back the data or packing version

Table 10. Agnes pretraining and release gates (continued).

Gate	Time	Pass criterion	Action on failure
G1	FP8 numerical path	At 170M and 600M, FP8–BF16 validation-loss difference ≤ 0.005 ; throughput $\geq 1.25\times$; no NaN or Inf	Return sensitive layers to BF16; recalibrate scaling and initialization
G2	Optimizer	Two sizes and at least two seeds; MuonH endpoint no worse than the tuned baseline; no anomalous effective-LR collapse	Retain AdamW or standard Muon as the production fallback
G3	Systems	24-GPU rate ≥ 220 and 96-GPU rate ≥ 175 TFLOP/s/GPU; stage imbalance $< 8\%$; memory headroom $\geq 1.5\text{GB}$	Adjust PP layout, MBS, and overlap; then recompute time and budget
G4	Phase 1	Validation loss ≤ 2.75 ; curve no more than 2% outside the 600M extrapolation band; no statistically material regression on Chinese or general proxies under the frozen test	Pause Phase 2 and inspect data weights, LR, and numerical execution
G5	Final capability	Validation loss ≤ 2.52 ; Avg15 ≥ 57.0 ; Math/Code Avg4 ≥ 42.0 ; Reason/Knowledge Avg11 ≥ 62.5 ; see Table 9	Compare uniform, curriculum, and averaged endpoints; select only a checkpoint that passes
G6	Release	Decontamination report, license manifest, reproducibility checklist, weight hashes, and model card approved; no P0 safety blocker	Do not release the failing checkpoint; retain an internal audit copy

Origin and status of the target values

The loss and aggregate capability thresholds in G4 and G5 are derived from endpoints reported at a similar model scale and training-token count, with a small engineering tolerance. They are **Agnes acceptance targets**, not performance promises. Per-task anchors in Table 9 are diagnostic and do not create hidden release gates. If tokenizer choice, licensing constraints, or stock-driver throughput changes the experimental conditions, reviewers may revise a threshold before production starts. Every revision must be versioned; no gate may be rewritten after final scores are observed.

8.4 Minimum Ablation Matrix

Table 11. Minimum experimental matrix before and during production.

Experiment	Scale / budget	Control	Primary criteria
Precision pairing	170M, 600M; 20 TPP	BF16 vs. blockwise FP8; same seed	Loss delta, throughput, saturation, memory
Optimizer pairing	170M, 600M; two seeds	AdamW, tuned Muon, MuonH	Endpoint loss, effective LR, stability
LR schedule	600M; 20/50 TPP	Grid over peak rate and decay ratio	Endpoint loss, extrapolation error
Systems scaling	1.7B; 1,000 steps	8/24/96 GPUs; candidate PP layouts	TFLOP/s, stage balance, AllReduce time
Data proxy	0.6B; source slices	Shared checkpoint plus replay	Four-axis vector, variance
Curriculum order	2B; Phase 2 shadow	Uniform vs. curriculum; identical tokens	Avg15, contamination, loss
Late branch	2B; final 29B	Decay final/average; constant-LR average	Stability, multi-endpoint capability

Curriculum ordering, constant learning rate, and model averaging do not have full-factorial evidence. The final report must therefore avoid attributing a joint improvement to any single component without a corresponding control. Under budget pressure, the uniform shadow and the three terminal comparators are retained because they determine whether to release the curriculum-averaged model or a more conservative single checkpoint.

9 Risk Analysis, Limitations, and Stop Conditions

9.1 Risk Register

Table 12. Principal technical and governance risks.

ID	Risk	Early signal	Prevention and response
R1	FP8 numerical divergence	Rising amax or saturation, loss spikes, non-finite gradients	Run paired BF16 controls; whitelist sensitive layers; pause automatically; restore a healthy checkpoint and fall back layerwise
R2	Consumer-GPU communication bottleneck	96-GPU rate below 175 TFLOP/s/GPU; increasing synchronization fraction	Use TP=1, topology-aware ranks, and asymmetric PP; retain stock drivers as the reproducible baseline
R3	Lack of ECC or node failure	GPU Xid events, silent corruption, temperature or power drift	Hold 10% spare GPUs; sample tensor hashes; exercise RPO/RTO procedures; isolate failing nodes automatically

Table 12. Principal technical and governance risks (continued).

ID	Risk	Early signal	Prevention and response
R4	Licensing or PII failure	Missing license, withdrawal request, anomalous PII hit rate	Enforce license whitelist and quarantine; retain deletable provenance; require legal approval before release
R5	Apparent curriculum gain caused by contamination	Rising near-duplicate rate between late buckets and benchmarks	Apply exact, MinHash, and embedding-based benchmark audits; report decontaminated reevaluation
R6	Misinterpreted source-local score	Higher-score buckets reduce proxy capability	Validate the semantic direction of every source score; otherwise use a fixed random order
R7	Token-ledger inconsistency	Materialized, step-derived, and non-padding counts differ by more than 0.1%	Freeze tokenizer first; maintain three ledgers; define late averaging by token count rather than step ID
R8	Proxy decisions fail to extrapolate	Model-size ordering changes or variance across seeds is high	Confirm critical choices at two scales; retain the conservative uniform alternative when disagreement persists
R9	Cost scope is misrepresented	USD 6.89K is described as total project cost	Separate accelerator-only, infrastructure, and labor accounting in every external document

9.2 Mandatory Stop Conditions

Training stops if any of the following occurs: validation loss worsens at two consecutive full validation points after excluding a scheduled data-transition effect; a non-finite parameter or gradient cannot be cleared by the automatic fallback path; licensing review identifies a P0 blocker; two consecutive checkpoints fail to restore; measured throughput would exceed the approved budget by more than 15% without a signed change request; or evaluation contamination cannot be repaired while preserving provenance.

9.3 Limitations

The plan has several material limitations. First, the principal efficiency evidence comes from shorter token-per-parameter regimes than the proposed production run; optimizer and curriculum rankings may change at approximately 700 tokens per parameter. Second, consumer GPUs lack ECC and NVLink, creating reliability and communication risks that accelerator-hour accounting alone does not capture. Third, the production tokenizer and several optimizer details are original Agnes choices rather than disclosed facts from the reference study. Fourth, the initial contamination policy cannot guarantee removal of semantic paraphrases or undisclosed benchmark derivatives. Fifth, the 40,960-token structural RoPE setting is not evidence of long-context training or performance; release evaluation remains capped at 4K. Finally, the proposed cost excludes material categories of project expenditure and depends on reproducing the measured throughput with supported software.

These limitations motivate the gate structure. Failure of an individual method does not require abandoning the program: BF16, AdamW or standard Muon, uniform sample order, stock drivers,

and single-checkpoint release remain conservative fallbacks. A fallback changes time, cost, or quality expectations and therefore requires a versioned planning update.

10 Reproducibility and Release Artifacts

10.1 Required Deliverables

The release candidate must contain:

1. **Agnes 2B Base:** BF16 SafeTensors, model configuration, generation configuration, and file-level SHA-256 hashes.
2. **Agnes tokenizer:** model files, special-token map, normalizer, tokenizer-training manifest, and fertility report.
3. **Data package:** source-and-license manifest, processing configuration, deduplication and decontamination reports, and phase- and bucket-level token ledgers. Restricted data are represented by reconstructable manifests and scripts rather than redistributed content.
4. **Training package:** container digest, Megatron configuration, parallel layouts, random seeds, full logs, key checkpoints, and recovery instructions.
5. **Evaluation package:** dataset versions, prompts, shot counts, decoding parameters, parsers, per-example outputs, and aggregation scripts.
6. **Model card and technical report:** intended scope, known limitations, cost boundary, data risks, base capabilities, and interfaces for subsequent training.

10.2 Pre-Release Checklist

Table 13. Reproducibility checklist. Every item remains pending until supported by project evidence.

Status	Item	Required evidence
☐	Raw-data versions and licenses locked	Source URL, version, license, access date, redistribution rule, and hashes
☐	Tokenizer fully frozen	Training manifest, vocabulary, normalizer, special tokens, and round-trip report
☐	Cleaning, deduplication, and contamination checks reproducible	Code revision, configuration, seeds, thresholds, output hashes, and deletion map
☐	Three token ledgers reconciled	Unique documents, materialized tokens, and physical non-padding tokens
☐	Training environment locked	Container, driver, CUDA, NCCL, framework, and kernel versions
☐	Model and optimizer configuration complete	Architecture, initialization, parameter groups, LR, precision, parallelism, and exact parameter count
☐	Checkpoint restoration demonstrated	Model, master weights, optimizer, scheduler, RNG, data cursor, and recovery-drill logs
☐	Evaluation version locked	Dataset versions, prompts, decoding, parsers, per-example outputs, and statistics scripts

Table 13. Reproducibility checklist (continued).

Status	Item	Required evidence
☐	Cost accounting auditable	Active GPU-hours, downtime, experiment/production split, rate assumptions, and exclusions
☐	Final artifacts verifiable	Weights and configuration hashes, model card, license manifest, and approval record

11 Conclusion

This report defines an auditable route to pretraining Agnes 2B rather than merely reproducing a low accelerator-cost headline. The plan combines consumer-GPU execution, blockwise FP8, explicitly scaled optimizer groups, source-local curriculum ordering, and late-stage parameter averaging within a single 1.3988T-token ledger. It decomposes the program into tokenizer and data freezes, scale-model experiments, two production phases, a uniform shadow branch, and multi-endpoint evaluation. It also closes common planning gaps in licensing, decontamination, recovery, and total-cost interpretation.

The performance profile is explicit and falsifiable. The final checkpoint must reach validation loss at most 2.52, Avg15 at least 57.0, Math/Code Avg4 at least 42.0, and Reason/Knowledge Avg11 at least 62.5 under the frozen Agnes protocol. The corresponding external capability anchors are 57.81, 43.50, and 63.02; the separate Phase 2 trajectory-loss anchor is 2.488. All individual benchmark values remain diagnostics rather than undisclosed release conditions.

The recommended decision is to authorize T0/D0 and the 170M–1.7B scaling ladder first. The 24-to-96-GPU production window should be committed only after G0–G3 pass. This structure ensures that if an externally reported method does not reproduce on Agnes data or under a supported driver, the team can adopt a validated conservative fallback before consuming the full 1.4T-token budget.

A Frozen Configuration Summary

Table 14. One-page summary of the proposed production configuration.

Item	Frozen value or rule
Architecture	28 layers, $d = 2048$, $d_{ff} = 6144$, Q/KV heads=16/8, head dimension=128, untied LM head
Tokenizer	Agnes byte-level BPE, $V = 151,936$; version and hashes frozen at T0
Sequence / batch	4,096 / GBS 1,536 / MBS 2; at most 6,291,456 physical tokens per step
Phase 1	438.84B tokens; 24 GPUs; TP/PP/DP=1/2/12; power LR; layout (18 10)
Phase 2	959.99B tokens; 96 GPUs; TP/PP/DP=1/4/24; linear LR; layout (9 9 9 1)
Precision	Linear E4M3 FP8; activation groups 1×128 ; weight blocks 128×128 ; remaining paths BF16/FP32
Optimizer	MuonH for selected matrices, WD 0 and $10 \times$ base LR; AdamW for remaining parameters, WD 0.1
Curriculum	43.87B-token transition; 376 source-local progress buckets; same-token uniform shadow
Late branch	Constant LR over approximately the final 29B tokens; equal model-parameter average of six terminal checkpoints
Checkpointing	Full state every 1,000 steps; lightweight model every 250; RPO $\leq 1,000$ steps
Performance	Final loss ≤ 2.52 ; Avg15 ≥ 57.0 ; Math/Code Avg4 ≥ 42.0 ; Reason/Knowledge Avg11 ≥ 62.5
Release	All G0–G6 gates pass; BF16 Base, tokenizer, manifests, code, logs, evaluation, and model card

B Separation of Reported Facts and Agnes Planning Values

Table 15. Final check against presenting planned values as achieved results.

Quantity	Epistemic status	Permitted use
238 / 192 TFLOP/s/GPU	Reported by the reference study	Systems anchors only; Agnes release gates are 220 / 175
2.730 / 2.488 validation loss	Reported by the reference study	Basis for G4/G5 targets; never claim Agnes has reached them
22,514 GPU-hours / USD 6.89K	Reported under a narrow accounting scope	Production anchor; experimental and rerun budgets remain separate
FP8 loss difference 0.0031–0.0039	Reported scaling evidence	Agnes revalidation threshold is at most 0.005
Avg4 43.50 / Avg11 63.02 / Avg15 57.81	Reported endpoints	Agnes G5 floors are 42.0 / 62.5 / 57.0; individual scores remain diagnostic
2.032B configuration, tokenizer, and recovery policy	Agnes design decisions	Generate project evidence through tensor export, T0, and recovery drills

C Token and Step Accounting

Table 16 records the accounting identities used by the scheduler and the release ledger. Displayed domain components are rounded to two decimals and displayed shares to one decimal; phase totals, rather than sums of rounded cells, are authoritative.

Table 16. Authoritative token-accounting identities.

Quantity	Value	Interpretation
Maximum tokens per step	6,291,456	1,536 sequences $\times$ 4,096 tokens
Phase 1 authoritative total	438.84B	Approximately 69,752 optimization steps
Phase 2 component pool	938.06B	Unique pool before Phase 1 replay
Phase 1 replay in Phase 2	Approximately 21.93B	Integral of the declining replay weight over the 43.87B transition
Phase 2 physical total	959.99B	Pool consumption plus replay
Full-run total	1.3988T	Phase 1 plus physical Phase 2 training tokens
Phase 2 steps	Approximately 152,586	Reconciled at runtime using the non-padding token counter

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